

CANDLE: Capacity-Aware Neurogenesis for Decentralized Federated Class-Incremental Learning-Based Intrusion Detection

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Abstract

Collaborative intrusion detection across independent networks is constrained by three practical conditions: traffic records cannot be centralized, local data are strongly non-independent and identically distributed, and the set of attack classes changes over time. Conventional federated intrusion detection systems keep raw traffic local but commonly depend on a coordinating server and are usually evaluated with a fixed label space. This paper proposes **CANDLE** (Capacity-Aware Neurogenesis for **D**ecentralized fed-erated class-incremental **L**earning), a server-free framework for class-incremental network intrusion detection. Each collaborator organizes its classifier into reserve, plastic, and consolidated neuron groups. New attack classes are learned through selectively activated capacity, whereas parameters associated with previously acquired knowledge are protected from incompatible updates. To coordinate learning without a central aggregator, collaborators exchange compact context descriptors for peer selection and subsequently share masked model updates only within compatible neighborhoods. A capacity controller determines whether unused neurons should be reused, additional lightweight capacity should be introduced, or low-importance capacity should be reclaimed. CANDLE also includes a context-routing mechanism for inference when the task identity of an incoming flow is unavailable. The framework is evaluated on CICIOT2023 with 100 non-IID collaborators over six sequential tasks. After the final task, CANDLE retains 22.49% cumulative accuracy, compared with 0.79% for the server-free Plexus baseline, while their average accuracies across tasks are comparable (43.45% and 43.32%, respectively). Although centralized reference methods remain stronger in absolute performance, the results show that CANDLE substantially improves late-stage knowledge retention under server-free and replay-free constraints.

Keywords: Intrusion detection system, federated learning, decentralized federated learning, class-incremental learning, neurogenesis, continual learning, context-aware aggregation.

1. Introduction

Machine-learning-based Intrusion Detection Systems (IDS) have become a core component of network defense against increasingly diverse attacks. However, no single organization observes enough attack traffic to train a detector that generalizes across the full threat landscape [1]. Federated Learning (FL) addresses this by letting multiple defenders collaboratively train a shared model by exchanging parameter updates rather than raw traffic, thereby preserving the confidentiality of sensitive logs and flow records [2]. Following the introduction of FedAvg [2], a large body of work has refined the aggregation rule to cope with statistical heterogeneity across participants [3, 4, 5], and these advances have been transferred to network security, yielding federated IDS that match centrally trained detectors while keeping data on premises [6, 1]. Nevertheless, the canonical FL formulation depends on a central coordinator that aggregates every update, which reintroduces

a single point of failure, a communication and trust bottleneck, and a scalability ceiling, properties that are particularly undesirable for security infrastructure that must remain available under adversarial pressure [7].

Decentralized Federated Learning (DFL) removes the central coordinator: clients exchange models directly with neighbors over a peer-to-peer communication graph and aggregate locally through gossip-style protocols [8, 7]. This eliminates the single point of failure and distributes communication cost, and decentralized FL has already been applied to intrusion detection with encouraging results [9]. To cope with the non-IID data that is the norm in security, where defenders monitor different networks and observe disjoint attack mixes, several clustered and neighbor-selection schemes group clients by the similarity of their data distributions or gradients so that only compatible peers are aggregated [10, 11, 12, 13]. These methods, however, measure compatibility from a static snapshot of the data or model and assume each client optimizes a fixed objective; they do not account for a client whose task, and hence whose plastic parameters, shift over time. A continually learning IDS instead needs a collaboration criterion defined over the learning state: which neurons remain plastic, which encode consolidated knowledge, and which attack classes have been acquired, rather

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than over raw-data or gradient similarity alone.

Intrusion detection is also a moving target. New malware families and attack techniques appear continuously, so each defender receives its data as a stream of tasks that introduce previously unseen classes over time [14]. Retraining the detector from scratch on every new task is computationally prohibitive, whereas fine-tuning only on recent data induces catastrophic forgetting, the abrupt loss of competence on earlier attack classes [15, 16]. The standard remedy in class-incremental learning is to replay stored exemplars of old classes [17], but retaining raw attack traffic conflicts with the privacy and data-retention constraints that motivate federation. Federated class-incremental learning (FCIL) has begun to combine these two demands: GLFC compensates for local and global forgetting through class-balanced gradients and a proxy server [18], TARGET distills the previous global model with synthetic data to avoid storing exemplars [19], and, closest to our setting, FL-IIDS and incremental federated IDS for IoT extend this idea to intrusion detection under streaming attack classes [20, 21]. Yet all of these rely on a central server or relay node to coordinate forgetting compensation and aggregation, which reinstates the single point of failure that decentralization is meant to remove. Consequently, a decentralized IDS must simultaneously (i) operate without a central server, (ii) learn an expanding set of attack classes from streaming tasks, and (iii) preserve knowledge of past attacks without replaying sensitive data, a combination that neither decentralized FL nor existing server-based federated class-incremental learning addresses.

To meet these coupled requirements we propose CANDLE (Capacity-Aware Neurogenesis for Decentralized federated cLass-incremental LEarning), a server-free framework for continual intrusion detection. CANDLE builds on neurogenesis and expansion-based continual learning [22, 23], which grows and partitions a network’s neurons by age: reserve neurons supply capacity for new tasks, young neurons learn the current task, and mature neurons are frozen to protect previously acquired knowledge, so that new attack classes are absorbed without replaying old traffic. To make this local mechanism collaborative in a decentralized setting, each client summarizes its learning state in a compact Context Capsule that bundles activation prototypes, neuron age masks, importance scores, and capacity statistics. Peers exchange these capsules to cluster and to aggregate selectively, shielding mature knowledge while sharing plastic knowledge only among context-compatible neighbors. A capacity-aware controller monitors neuron availability and domain shift to decide, per task, whether to reuse, expand, or recycle capacity, preventing the exhaustion that limits neurogenesis-based learners under long task sequences.

The main contributions of this work are as follows:

- A replay-free decentralized FCIL framework for IDS. We formulate intrusion detection as a decentralized federated class-incremental learning problem and introduce CANDLE, which couples age-based neurogenesis with peer-to-peer collaboration so that defenders continually learn new attack classes without a central server and without storing raw attack traffic.
- Context-aware decentralized clustering. We design the Context Capsule, a lightweight descriptor of each client’s activation prototypes, age masks, neuron importance, reliability, and capacity usage, together with a similarity measure over these capsules that groups clients by what they have learned. This restricts collaboration to compatible peers and yields a collaboration topology that is robust to the non-IID attack distributions typical of decentralized IDS.
- Age-aware robust aggregation with capacity-aware neurogenesis. We propose a masked aggregation rule that shares only plasticity-region and context-compatible parameters, protecting mature neurons that encode old attack knowledge, together with a Capacity-Aware Neurogenesis Controller (CANC) that expands capacity through sparse adapters under domain shift and recycles low-importance neurons when reserves are depleted, jointly mitigating catastrophic forgetting and mis-clustered aggregation.
- Comprehensive empirical validation. We evaluate CANDLE on the large-scale CICIOT2023 benchmark with 100 non-IID clients under both a stationary and a streaming six-task setting. Among server-free methods, CANDLE achieves the strongest resistance to catastrophic forgetting, retaining accuracy on earlier attack classes where decentralized and regularization-based baselines collapse, while restricting communication to context clusters; centralized replay-based methods, which relax CANDLE’s server-free and replay-free constraints, serve as an upper reference.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 formalizes the problem setting, Section 4 details CANDLE, Section 5 reports the experimental evaluation, and Section 6 concludes.

2. Related work

This section reviews the three research directions that define the scope of CANDLE: FL-based intrusion detection, decentralized federated learning for IDS, and federated class-incremental intrusion detection. The discussion emphasizes the assumptions and limitations that affect the design of a server-free, replay-free, and temporally adaptive detector.

2.1. Federated Learning-based Intrusion Detection

FL-based IDS allows multiple organizations or edge domains to learn from private traffic without transferring raw records to a common repository. Most studies adopt a server-coordinated protocol derived from FedAvg, in which local models are optimized independently and combined according to the amount of data held by each participant [2]. Subsequent methods reduce client drift or improve aggregation under heterogeneous data through proximal regularization, momentum, or generalized fusion rules [3, 4, 5]. Surveys of federated cybersecurity systems show that this paradigm has been applied to IoT, industrial, enterprise, and vehicular intrusion detection, often with

detection performance close to centralized learning when the training distribution is stable [6, 1].

FedKD-IDS illustrates a different design choice within centralized FL-based IDS. Instead of exchanging full model parameters, participants generate predictions on a shared unlabeled dataset, and server-side verification and voting are used to filter suspicious contributions before distillation [24]. This design addresses communication cost, limited labels, non-IID data, and poisoning attacks. Nevertheless, the verification and fusion procedures remain dependent on a central server, and the output space is fixed during training. FedKD-IDS is therefore relevant to CANDLE as an example of how learning-state summaries can support robust collaboration, but it does not address server-free coordination or continual expansion of the attack label space.

Overall, FL-based IDS research establishes the value of collaborative learning with local data ownership, but it usually assumes that a central entity is available and that the detector is trained for a stationary set of classes. These assumptions motivate the decentralized and incremental directions considered next.

2.2. Decentralized Federated Learning-based Intrusion Detection

DFL replaces the aggregation server with direct communication over a graph. Each participant combines information from its neighbors and performs local training, which improves fault tolerance and removes a centralized coordination bottleneck [8, 7]. Decentralized online IDS studies demonstrate that peer-to-peer learning can support collaborative detection without a server [9]. However, the quality of a decentralized model depends heavily on the communication topology because every participant constructs its own aggregate from a local neighborhood.

Several methods therefore select or group participants according to validation loss, model preference, gradient correction, or consensus-based clustering [10, 11, 12, 13]. These methods reduce negative transfer under non-IID data, but their compatibility measures are primarily designed for a stationary learning objective. In a class-incremental IDS, two participants may appear similar during one communication round while preserving different histories of attack classes. A neighborhood rule that ignores this history can mix parameters that have different functional roles.

XDFC-IDS extends decentralized learning to an incremental IDS by combining a peer-to-peer mesh with regularization-based continual learning [25, 16]. Its results show the feasibility of server-free incremental training on streaming IoT traffic. At the same time, a dense mesh increases communication with the number of participants, and uniform aggregation does not explicitly distinguish peers according to their learned class history or remaining plastic capacity. CANDLE addresses this issue by making model maturity and class-level context part of the neighbor-selection process. XDFC-IDS is discussed only in this related-work context because it is not a direct experimental baseline in the present study.

2.3. Federated Class-Incremental Intrusion Detection

CIL requires a single classifier to recognize all classes observed up to the current task without receiving a task identifier at prediction time [14]. Catastrophic forgetting is commonly mitigated through replay, regularization, distillation, or architectural expansion. Replay-based methods preserve representative samples or synthetic surrogates from earlier tasks, whereas regularization-based methods constrain changes to parameters considered important for prior knowledge [17, 16]. Architectural methods instead allocate additional capacity to new tasks and protect previously consolidated components [22, 23].

FCIL introduces an additional source of forgetting because locally degraded models are repeatedly propagated through federated aggregation. GLFC compensates for local and global forgetting using class-aware optimization and server-supported reconstruction, while TARGET uses distillation with generated data to transfer knowledge from a previous global model [18, 19]. IDS-specific studies such as FL-IIDS and incremental federated IDS frameworks apply similar principles to evolving attack classes [20, 21]. These approaches improve retention, but they continue to rely on centralized coordination or historical-data surrogates.

For a decentralized IDS, the continual-learning mechanism must also be compatible with peer-specific model states. A participant that has consolidated one set of attacks should not overwrite that knowledge with parameters learned from a disjoint task stream. Existing FCIL methods do not explicitly represent neuron maturity or residual capacity when selecting peers. This limitation leads to the architectural perspective adopted by CANDLE.

2.4. Architectural Continual Learning and Research Gap

Dynamically Expandable Networks and NICE demonstrate that continual learning can be supported by allocating new capacity and protecting previously learned components [22, 23]. NICE is particularly relevant because it distinguishes unused, recently trained, and consolidated neurons and includes a context-based inference mechanism. These methods, however, were developed for a single learner. They do not define how independently evolving neuron groups should be aligned, compared, and aggregated across decentralized participants.

The literature consequently leaves a four-part gap. First, server-free IDS methods mainly assume a stationary label space. Second, FCIL-based IDS methods mainly retain centralized coordination. Third, decentralized neighbor-selection methods do not model the temporal maturity of learned parameters. Fourth, architectural continual-learning methods do not manage collaboration among multiple learners. CANDLE is designed at this intersection: it combines age-partitioned capacity, context-aware peer selection, masked decentralized aggregation, and task-agnostic inference in a unified IDS framework.

3. Background

This section defines the three learning settings underlying CANDLE. FL-IDS specifies how geographically distributed

traffic owners optimize a common detector. DFL-IDS changes the coordination topology by removing the central aggregator. FCIL-IDS changes the temporal learning objective by allowing new attack classes to appear over a sequence of stages. CANDLE operates at the intersection of the latter two settings.

Table 1: Notation used to describe FL-IDS, DFL-IDS, and FCIL-IDS.

Symbol	Description
$\mathcal{V} = \{1, \dots, M\}$	Set of participating monitoring sites
$q \in \{0, \dots, Q-1\}$	Communication cycle
$s \in \{1, \dots, S\}$	Incremental stage
$\mathcal{B}_i^s$	Private traffic batch of site i at stage s
$\mathcal{C}_i^s$	Classes introduced locally at stage s
$\mathcal{C}_i^{\leq s}$	Classes observed locally up to stage s
$\theta_i^{s,q}$	Parameters maintained by site i
$h(\cdot; \theta)$	Intrusion classifier
$\Gamma_i^{s,q}$	Communication neighborhood of site i
$\mathbf{A}^{s,q} = [a_{ij}^{s,q}]$	Decentralized mixing matrix
$\mathcal{M}_i^{s-1}$	Compact memory/state retained from earlier stages

3.1. Federated Learning-based Intrusion Detection

Consider M monitoring sites, such as IoT gateways, enterprise domains, or edge security nodes. Site i owns a private traffic collection $\mathcal{B}_i = \{(\mathbf{x}_{i,u}, c_{i,u})\}_{u=1}^{m_i}$, where $\mathbf{x}_{i,u}$ is a flow representation and $c_{i,u}$ is a benign or attack label. The local empirical risk is

$$J_i(\theta) = \frac{1}{m_i} \sum_{u=1}^{m_i} \ell(h(\mathbf{x}_{i,u}; \theta), c_{i,u}), \quad (1)$$

where $\ell(\cdot)$ is a classification loss. A centralized FL-IDS seeks a model that balances the local objectives,

$$\theta^* \in \arg \min_{\theta} \sum_{i=1}^M \omega_i J_i(\theta), \quad \omega_i \geq 0, \quad \sum_{i=1}^M \omega_i = 1. \quad (2)$$

The coefficients ω_i may reflect local sample counts, data quality, or a participation policy. During communication cycle q , the server distributes a reference state $\bar{\theta}^q$. Each site performs local optimization,

$$\theta_i^{q+1} = \text{LOCALTRAIN}(\bar{\theta}^q, \mathcal{B}_i), \quad (3)$$

and the server constructs the next reference state from the submitted local models,

$$\bar{\theta}^{q+1} = \sum_{i=1}^M \omega_i \theta_i^{q+1}. \quad (4)$$

In an IDS, the local distributions are rarely identical. Different sites observe different protocols, devices, benign workloads, and attack prevalence. As a result, Eq. (2) represents a compromise among heterogeneous security environments rather than repeated samples from one common distribution. The central server also becomes operationally critical because all training cycles depend on Eqs. (3) and (4). DFL-IDS removes this dependency.

3.2. Decentralized Federated Learning-based Intrusion Detection

A DFL-IDS connects the monitoring sites through a graph $\mathcal{G}^{s,q} = (\mathcal{V}, \mathcal{E}^{s,q})$. Site i communicates only with the neighborhood $\Gamma_i^{s,q}$ and does not receive a server-generated global model. Let $\mathbf{A}^{s,q}$ be a row-stochastic mixing matrix, where $a_{ij}^{s,q} > 0$ only when $j = i$ or $j \in \Gamma_i^{s,q}$ and $\sum_j a_{ij}^{s,q} = 1$. A generic consensus-then-adapt cycle is

$$\bar{\theta}_i^{s,q} = \sum_{j=1}^M a_{ij}^{s,q} \theta_j^{s,q}, \quad (5)$$

$$\theta_i^{s,q+1} = \bar{\theta}_i^{s,q} - \eta \widehat{\nabla} J_i^s(\bar{\theta}_i^{s,q}), \quad (6)$$

where $\widehat{\nabla} J_i^s$ is a stochastic gradient computed from local traffic at stage s .

The mixing matrix determines which security environments influence one another. A dense matrix increases communication and may combine incompatible models, whereas a sparse matrix reduces traffic but can slow information propagation. In a stationary DFL problem, $\mathbf{A}^{s,q}$ can be fixed or selected from current losses or gradients. In an incremental IDS, however, compatibility must also reflect class history and parameter maturity. CANDLE therefore treats the graph as task-adaptive and derives peer relationships from compact learning-state information.

3.3. Federated Class-Incremental Intrusion Detection

In FCIL-IDS, site i receives a sequence of private traffic batches $\mathcal{B}_i^1, \dots, \mathcal{B}_i^S$. The classes introduced at stage s are $\mathcal{C}_i^s$, and the cumulative local label space is

$$\mathcal{C}_i^{\leq s} = \bigcup_{v=1}^s \mathcal{C}_i^v. \quad (7)$$

The detector evaluated after stage s must recognize every class in $\mathcal{C}_i^{\leq s}$ without being told which stage generated the test sample. A general local objective can be written as

$$\min_{\theta_i^s} J_i^s(\theta_i^s) + \lambda \mathcal{R}_i^s(\theta_i^s; \mathcal{M}_i^{s-1}), \quad (8)$$

where the first term learns the current traffic batch and $\mathcal{R}_i^s$ preserves earlier knowledge using a compact state $\mathcal{M}_i^{s-1}$. Depending on the method, $\mathcal{M}_i^{s-1}$ may contain exemplars, a previous model, importance scores, prototypes, or an architectural mask. Replay methods use historical samples; regularization and distillation methods constrain functional or parameter drift; architectural methods allocate protected capacity to earlier stages.

Forgetting should be measured on earlier stages rather than inferred only from the current aggregate accuracy. Let $A_i(s, v)$ denote the accuracy, after training through stage s , on the test set associated with stage v . The mean backward performance loss at site i is

$$\mathcal{F}_i(s) = \frac{1}{s-1} \sum_{v=1}^{s-1} [A_i(v, v) - A_i(s, v)], \quad s > 1. \quad (9)$$

A larger value indicates stronger forgetting. In a federated setting, a locally forgotten model can influence other participants through aggregation, so stability must be controlled both within each site and across the communication topology.

3.4. Joint DFL-FCIL Setting for IDS

Combining Eqs. (5)–(8), a decentralized class-incremental IDS can be viewed as the repeated operator

$$\theta_i^{s,q+1} = \mathcal{U}_i^s \left(\sum_{j=1}^M a_{ij}^{s,q} \theta_j^{s,q}; \mathcal{B}_i^s, \mathcal{M}_i^{s-1} \right), \quad (10)$$

where $\mathcal{U}_i^s$ combines local incremental learning with preservation of past knowledge. This formulation exposes two coupled design questions. First, what information should be retained locally in $\mathcal{M}_i^{s-1}$ to balance stability and plasticity without storing sensitive historical traffic? Second, how should $a_{ij}^{s,q}$ be chosen so that a participant is influenced only by peers whose learned context and trainable capacity are compatible? CANDLE addresses the first question through age-partitioned neurogenesis and the second through Context-Capsule-based neighborhood construction and masked aggregation.

3.5. Neurogenesis-based Continual Learning

Architectural continual learning preserves old knowledge by assigning different functional roles to different parts of a network. Dynamically Expandable Networks introduce new capacity when the existing representation is insufficient and prune redundant connections after adaptation [22]. NICE organizes neurons by learning age so that unused capacity can be activated for a new task and previously consolidated capacity can be protected from further modification [23]. This strategy is attractive for IDS because it does not require a replay buffer of raw traffic. Its main limitations in a decentralized federation are parameter alignment, finite reserve capacity, and the absence of a rule for deciding which participants should exchange which trainable components. These limitations motivate the CANDLE mechanisms presented in Section 4.

4. Methodology

4.1. Overview and Problem Formulation

As motivated in Section 1, a decentralized intrusion detector must satisfy three requirements simultaneously that no single prior method covers: it must operate without a coordinating server, it must absorb a sequence of tasks that introduce previously unseen attack classes, and it must do so without ever sharing or reconstructing raw attack traffic across collaborators. Restated more formally, let $\mathcal{N} = \{1, \dots, N\}$ be the set of collaborators, and let $\mathcal{T} = \{1, \dots, T\}$ index a sequence of tasks such that, at task t , every collaborator n receives a private dataset $\mathcal{D}_n^t$ whose label set $\mathcal{Y}_n^t$ may contain classes never seen by n before, possibly none of which overlap with the classes newly observed by another collaborator $k \neq n$ in the same task. Collaborators communicate over a graph $\mathcal{G}^t = (\mathcal{N}, \mathcal{E}^t)$ that, unlike the static $\mathcal{P}_n$ of Section 3.2, is allowed to be re-derived at every task from

what each collaborator has actually learned. No server, proxy, or relay node is assumed to exist at any point of the pipeline, and no collaborator is permitted to transmit raw samples, reconstructed exemplars, or synthetic data standing in for $\mathcal{D}_n^t$, $t' < t$. The learning objective for every collaborator n at task t is therefore

$$\min_{w_n^t} \underbrace{\mathcal{L}_{\text{Clf}}(f_{w_n^t}(X_n^t), Y_n^t)}_{\text{new-task fit}} + \underbrace{\Omega(w_n^t, w_n^{t-1})}_{\text{old-task protection}} \quad \text{s.t. } \Xi_n^t \subseteq \{\kappa_k^t : k \in \mathcal{N} \setminus \{n\}\}, \quad (11)$$

where $\Omega(\cdot, \cdot)$ must protect the classes in $\mathcal{Y}_n^{t-1}$ without reading $\mathcal{D}_n^{t-1}$, and Ξ_n^t denotes the compact, capsule-only information collaborator n is allowed to exchange with peers, formalized below. The generic objective above does not determine how previous knowledge is protected or how peer influence is restricted when task histories diverge. CANDLE resolves these choices through age-partitioned capacity and learning-state-aware neighborhood construction, as described below.

CANDLE at a glance. We address Eq. (11) by restructuring both the local model and the aggregation rule around the notion of *neuron age*, following the neurogenesis-based continual learning of DEN [22] and NICE [23], and by making that age structure legible to peers through a compact summary that replaces both raw data and the uniform EWC vector of Eq. (??). Four mechanisms compose the framework, illustrated in Fig. 1 and detailed step by step in Section 4.3:

1. *Age-based neurogenesis backbone.* Every collaborator’s C parameters are partitioned by an age mask $a_n^t \in \{\mathcal{R}, \mathcal{Y}, \mathcal{M}\}^C$ into a *reserve* pool $\mathcal{R}_n$ (untrained capacity held for future tasks), a *young* pool $\mathcal{Y}_n$ (plastic parameters actively trained on the current task), and a *mature* pool $\mathcal{M}_n$ (frozen parameters consolidated from earlier tasks). Only $\mathcal{Y}_n$ receives gradient updates at task t , so old-class knowledge in $\mathcal{M}_n$ is protected structurally rather than penalized as in Eq. (??), and no exemplar buffer is required.
2. *Context Capsule.* At the end of local training, collaborator n summarizes its learning state into a lightweight capsule κ_n^t bundling class-wise activation prototypes, the current age mask, per-parameter importance scores, and a capacity/reliability scalar. The capsule is the only artifact ever transmitted in place of raw data or full EWC vectors.
3. *Context-aware, age-masked aggregation.* Collaborators exchange capsules and cluster themselves into a task-adaptive neighbor set $\mathcal{N}_n^t$ by capsule similarity; aggregation is then restricted to the young pool of context-compatible neighbors, so a peer whose current task is unrelated to n ’s never overwrites n ’s plastic parameters, and the mature pool $\mathcal{M}_n$ is excluded from aggregation altogether.
4. *Capacity-Aware Neurogenesis Controller (CANC).* At the end of every task, a per-client controller inspects reserve utilization and prototype drift to decide whether to reuse the current young pool, expand it by promoting additional reserve neurons or sparse adapters, or recycle low-importance mature neurons back into reserve, preventing the capacity exhaustion that unmanaged neurogenesis would otherwise suffer over a long task sequence.

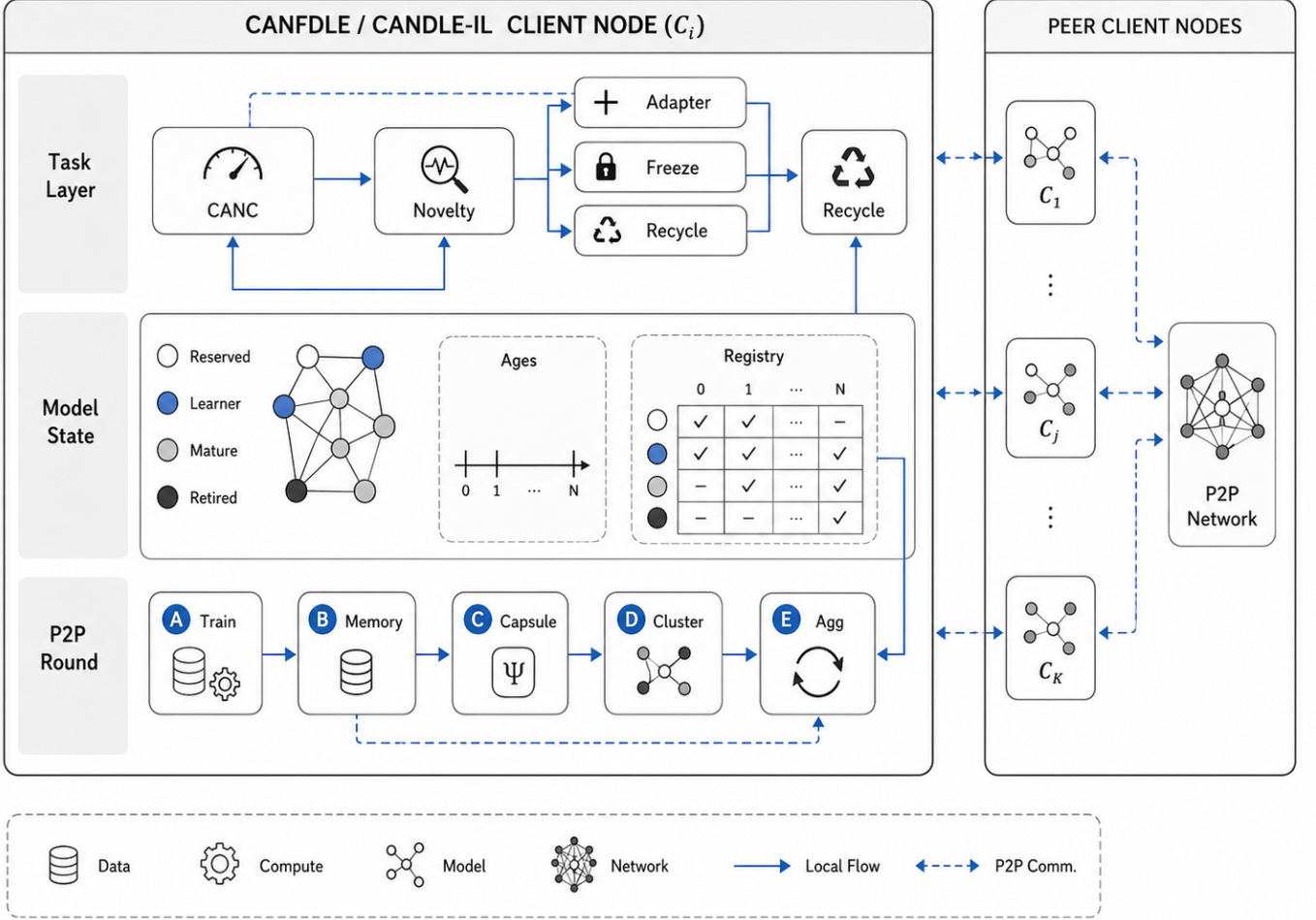


Figure 1: Overview of the proposed CANDLE workflow: local age-partitioned learning, Context Capsule construction and exchange, context-aware masked aggregation, and capacity-aware neurogenesis control.

Table 2 extends the notation of Table 1 with the symbols specific to CANDLE.

Table 2: Additional notations introduced by CANDLE.	
Notation	Description
$a_n^t \in \{\mathcal{R}, \mathcal{Y}, \mathcal{M}\}^C$	Age mask of collaborator n at task t
$\mathcal{R}_n, \mathcal{Y}_n, \mathcal{M}_n$	Reserve, young, mature index sets
$p_n^t(y)$	Activation prototype of class y for collaborator n
$\kappa_n^t = (P_n^t, a_n^t, F_n^t, \rho_n^t)$	Context Capsule of collaborator n at task t
ρ_n	Capacity/reliability scalar of collaborator n
ξ	Capsule-similarity threshold
$\mathcal{N}_n^t$	Context-compatible neighbor set of n at task t
u_n^t, δ_n^t	Reserve utilization, prototype drift score
θ_1, θ_2	CANC decision thresholds

4.2. System and Threat Assumptions

CANDLE considers a consortium of authenticated monitoring sites connected through reliable peer-to-peer channels. Participants may hold heterogeneous and non-IID traffic distributions, and their local class sets may evolve asynchronously. The current formulation assumes that collaborating sites execute the prescribed training and aggregation protocol; Byzantine model poisoning, Sybil participation, and communication-layer impersonation are outside the evaluated threat model. The sites use a compatible backbone architecture and a shared parameter-indexing scheme for coordinates that may be aggregated. Newly discovered attack classes are assumed to be labeled locally, for example through analyst validation, before they are introduced as an incremental stage.

Raw traffic records and labels are not transmitted. Nevertheless, CANDLE exchanges context descriptors and masked model updates, which may reveal information under model-inversion or membership-inference attacks. Accordingly, the present work claims data locality rather than a formal privacy guarantee. Differential privacy, secure aggregation, and Byzantine-robust filtering can be integrated in future extensions but are not assumed in the current evaluation.

4.3. Detailed Workflow of CANDLE

For task t , every collaborator n executes the seven steps below; Steps 2-5 are repeated over R communication rounds, while Steps 1, 6, and 7 execute once per task, mirroring the round/task structure of Section ??.

Step 1 – Task Acquisition and Reserve Activation.. At the first round of task t , collaborator n receives $\mathcal{D}_n^t$ and compares $\mathcal{Y}_n^t$ against $\mathcal{Y}_n^{t-1}$. If new classes are present, a Promote($\cdot$) operator converts $k_n^t = \kappa \cdot |\mathcal{Y}_n^t \setminus \mathcal{Y}_n^{t-1}|$ reserve neurons into young neurons, where κ is a fixed capacity-per-class budget,

$$a_n^{t,1} = \text{Promote}(a_n^{t-1}, \min(k_n^t, |\mathcal{R}_n^{t-1}|)). \quad (12)$$

If $\mathcal{Y}_n^t = \mathcal{Y}_n^{t-1}$ (no new class for n at task t), the mask is carried over unchanged and only the mature-anchored regularization of Step 2 is applied.

Step 2 – Local Age-Masked Dual-Loss Learning.. For E local epochs, collaborator n updates only its young pool with a classification loss identical in form to Eq. (??) but masked by a_n^t ,

$$w_n^{r,t} = w_n^{r-1,t} - \alpha \mathcal{H}[a_n^t = \mathcal{Y}] \odot \nabla_w \mathcal{L}_{\text{Clf}}(w_n^{r-1,t}), \quad (13)$$

and regularizes the young pool toward the mature-consolidated reference w_n^{temp} using an importance-weighted penalty restricted to the mature index set,

$$\mathcal{L}_{\text{Reg}}(w_n^{r,t}, w_n^{\text{temp}}) = \frac{\lambda}{2} \sum_{c \in \mathcal{M}_n^t} F_{n,c}^{t-1} (w_{n,c}^{r,t} - w_{n,c}^{\text{temp}})^2. \quad (14)$$

Because Eq. (14) sums only over $\mathcal{M}_n^t \subset \{1, \dots, C\}$ rather than the full parameter vector as in Eq. (??), the regularization cost of CANDLE does not grow uniformly with t ; it grows only with the size of the pool that has actually been consolidated.

Step 3 – Context Capsule Construction.. After local training, collaborator n computes, on its local evaluation split, one activation prototype per known class,

$$p_n^t(y) = \frac{1}{|\mathcal{I}_n^t(y)|} \sum_{i \in \mathcal{I}_n^t(y)} g_{w_n^{R,t}}(X_{n,i}), \quad \mathcal{I}_n^t(y) = \{i : y_{n,i} = y\}, \quad (15)$$

where $g_w(\cdot)$ denotes the penultimate-layer embedding of the classifier, refreshes the importance scores $F_{n,c}^t$ over the young pool using the diagonal Fisher approximation of Eq. (??), and computes a capacity scalar $\rho_n^t = |\mathcal{R}_n^t|/C$. These three quantities, together with the current age mask, are packaged into

$$\kappa_n^t = (P_n^t, a_n^t, F_n^t, \rho_n^t), \quad P_n^t = \{p_n^t(y)\}_{y \in \mathcal{Y}_n^t}. \quad (16)$$

No entry of X_n^t or y_n^t appears in κ_n^t ; the capsule is bounded in size by $|\mathcal{Y}_n^t| \cdot d + C$ regardless of $|\mathcal{D}_n^t|$.

Step 4 – Capsule Exchange and Context-Aware Clustering.. Collaborator n broadcasts κ_n^t to $\mathcal{N} \setminus \{n\}$ (or to a candidate set reachable within a bounded number of hops) and receives $\{\kappa_k^t\}_{k \neq n}$ in return. Pairwise compatibility combines prototype agreement on shared classes with age-mask overlap, in the spirit of the Hamming-distance verification used server-side by FedKD-IDS [24] but computed peer-to-peer and over learning state rather than logits,

$$\text{sim}(\kappa_n, \kappa_k) = \beta \cdot \overline{\cos}(P_n^t, P_k^t) + (1 - \beta) \cdot (1 - \text{HD}(a_n^t, a_k^t)), \quad (17)$$

where $\overline{\cos}(\cdot, \cdot)$ averages cosine similarity over classes in $\mathcal{Y}_n^t \cap \mathcal{Y}_k^t$ and $\text{HD}(\cdot, \cdot)$ is the normalized Hamming distance between age masks. The context-compatible neighbor set is then

$$\mathcal{N}_n^t = \{k \in \mathcal{N} \setminus \{n\} : \text{sim}(\kappa_n^t, \kappa_k^t) \geq \xi\}. \quad (18)$$

Step 5 – Age-Aware Masked Peer Aggregation.. Collaborator n aggregates only the young-pool parameters of context-compatible neighbors, weighted by dataset size and by the reliability scalar ρ_k^t ,

$$w_n^{r,t}[\mathcal{Y}_n] \leftarrow w_n^{\text{pre}}[\mathcal{Y}_n] - \sum_{k \in \mathcal{N}_n^t \cup \{n\}} \frac{|\mathcal{D}_{k,T}| \rho_k^t}{\sum_{j \in \mathcal{N}_n^t \cup \{n\}} |\mathcal{D}_{j,T}| \rho_j^t} \Delta w_k^{r,t}[\mathcal{Y}_n \cap \mathcal{Y}_k], \quad (19)$$

while $w_n^{r,t}[\mathcal{M}_n]$ is left untouched. Equation (19) specializes Eq. (??) in two ways: the summation ranges over $\mathcal{N}_n^t$ instead of a fixed peer set, and the update is applied coordinate-wise only where the receiving and sending parameters are both young and correspond to overlapping classes, so a peer that has fully drifted onto disjoint classes contributes nothing even if it passes the similarity threshold of Eq. (18). Steps 2-5 repeat for R rounds to complete task t .

Step 6 – Capacity-Aware Neurogenesis Control. At the end of task t ($r = R$), the local CANCE computes reserve utilization and a domain-shift score measuring how far the current prototypes have moved from the previous task’s,

$$u_n^t = 1 - \frac{|\mathcal{R}_n^t|}{C}, \quad \delta_n^t = \|\bar{P}_n^t - \bar{P}_n^{t-1}\|_2, \quad (20)$$

where $\bar{P}_n^t$ averages Eq. (15) over classes common to tasks $t-1$ and t . CANCE then selects one of three actions:

$$\text{CANCE}(u_n^t, \delta_n^t) = \begin{cases} \text{REUSE}, & u_n^t < \theta_1 \text{ and } \delta_n^t < \theta_2, \\ \text{EXPAND}, & (u_n^t \geq \theta_1 \text{ or } \delta_n^t \geq \theta_2) \text{ and } |\mathcal{R}_n^t| > 0, \\ \text{RECYCLE}, & |\mathcal{R}_n^t| = 0. \end{cases} \quad (21)$$

REUSE carries the current young pool into task $t+1$ unchanged; EXPAND promotes additional reserve neurons, or, if $\delta_n^t \geq \theta_2$ indicates a sharp domain shift rather than a simple new-class arrival, attaches sparse adapters to the young pool; RECYCLE identifies the mature neurons with the lowest importance, $\{c \in \mathcal{M}_n^t : F_{n,c}^t \text{ percentile}_{\theta_3}(F_n^t)\}$, and reinitializes them back into $\mathcal{R}_n^t$, trading a small amount of old-class fidelity for renewed capacity so that neurogenesis does not stall over long task sequences, the failure mode left unaddressed by DEN and NICE in Section 2.3.

The capacity pressure that triggers each action is computed per layer as

$$\kappa_{n,l}^t = \alpha(1 - \rho_{n,l}^0) + \beta u_{n,l}^{t-1} + \gamma \Delta L_{\text{val},n}^t + \delta \delta_n^t, \quad (22)$$

where $\rho_{n,l}^0 = |\{c \in l : a_{n,c}^t = \mathcal{R}\}|/|l|$ is the per-layer reserve fraction, $u_{n,l}^{t-1}$ measures the neuron consumption of the preceding task at layer l , $\Delta L_{\text{val},n}^t$ captures the change in validation loss from the previous task, and $\alpha, \beta, \gamma, \delta$ are mixing coefficients. When EXPAND is selected and $\delta_n^t \geq \theta_2$ indicates a sharp domain shift rather than a mere new-class arrival, CANCE attaches a lightweight adapter module to the young pool:

$$z_l = f_l(z_{l-1}) + A_{n,l}^t(z_{l-1}), \quad (23)$$

where $A_{n,l}^t$ is a low-rank linear layer whose parameters are young by construction and therefore included in the age-masked aggregation of Step 5. Adapter parameters are aggregated separately from the backbone parameters, only when the adapter key and shape match between sender and receiver, preventing architectural mismatches from corrupting the aggregation step.

Step 7 – Consolidation and Deployment. The young neurons trained during task t transition to mature, $a_n^{t+1} \leftarrow \text{Consolidate}(a_n^t)$, the importance scores F_n^t are finalized as in Eq. (??), and $w_n^{\text{temp}} \leftarrow w_n^{R,t}$ is stored as the anchor for Eq. (14) in task $t+1$. Collaborator-specific parameters $w_n^{R,t}$ are then deployed for online detection, and the collaborator awaits task $t+1$, at which point $w_n^{R,t}$ serves as the initialization for continued decentralized training, with newly joined collaborators entering the federation directly at Step 4 once they broadcast their first capsule.

4.4. Context-Aware Inference

At deployment time, collaborator n classifies an incoming network flow x without access to a task identifier. The inference procedure exploits the context memory accumulated during Step 2 of each training round (Section 4.3) and proceeds in three stages:

1. **Activation extraction and binarization.** The classifier $f_{w_n}(\cdot)$ performs a forward pass on x , and the activations at each intermediate layer l are binarized using the thresholds calibrated during task 0:

$$b_l(x) = \mathbb{I}[\text{act}_l(x) > \mu_l + \sigma_l], \quad (24)$$

where μ_l and σ_l are the per-layer mean and standard deviation of the activations collected on the first task.

2. **Episode routing.** The binarized vector $b(x) = [b_1(x); \dots; b_L(x)]$ is compared against the stored activation memories $\{m_n^{(e)}\}_{e=1}^{|\mathcal{T}_n|}$, one per learned episode (task), and the best-matching episode is selected:

$$\hat{e} = \arg \max_{e \in \mathcal{T}_n} \text{sim}(b(x), m_n^{(e)}), \quad (25)$$

where $\text{sim}(\cdot, \cdot)$ is a binary pattern similarity (e.g., Hamming or cosine on the binary vectors). This routing mechanism is inherited from NICE [23] but is now executed per-client in a decentralized setting, with each collaborator maintaining its own context detector without server coordination.

3. **Masked classification.** The output logits are masked to retain only the classes associated with episode $\hat{e}$, and the final prediction is:

$$\hat{y} = \arg \max_{y \in \mathcal{Y}_n^{(\hat{e})}} f_{w_n}(x)[y]. \quad (26)$$

By restricting the output space to the classes of the routed episode, context-aware inference reduces inter-task confusion that would otherwise grow as the cumulative label space $|\bigcup_t \mathcal{Y}_n^t|$ increases. It is worth noting that this mechanism does not detect truly novel, previously unseen attack classes; unknown traffic is forwarded to the Security Operations Center (SOC) for labeling and incorporated as a new incremental task in subsequent training rounds.

5. Experiment

We evaluate CANDLE on a large-scale IoT intrusion-detection benchmark under two scenarios: a stationary setting that verifies the model can fit a fixed data distribution, and a streaming setting that measures how well it learns an expanding set of attack classes without forgetting. All methods are trained on the same client partition and evaluated on a held-out global test set.

5.1. Research Questions

Our evaluation is organized around three research questions (RQs):

- **RQ1 (Stationary capability).** Operating without a central coordinator and without replaying past data, can CANDLE learn an effective intrusion detector on a fixed (stationary) attack distribution?
- **RQ2 (Continual retention).** In the streaming setting where new attack classes arrive over time, how well does CANDLE resist catastrophic forgetting relative to server-free and server-based baselines?
- **RQ3 (Communication efficiency).** How does CANDLE’s per-round communication cost compare with that of decentralized and federated baselines?

5.2. Dataset

We use CICIOT2023 [26], a real-time IoT traffic dataset comprising benign traffic and 33 attack types (34 classes in total) spanning DDoS, DoS, reconnaissance, spoofing, web, brute-force, and Mirai families. We split the 34 classes into six sequential tasks: the first five tasks introduce six classes each and the last task introduces the remaining four. The training data is distributed over 100 clients under a non-IID Dirichlet partition with concentration $\alpha = 5.0$, which yields moderate label skew across clients. Figure 2 visualizes the resulting per-client label distribution.

5.3. Experimental Scenarios

Scenario 1 (stationary). The first scenario isolates the base task $t=0$, whose six classes are fixed throughout training. Every client trains only on its local shard of this task and the model is evaluated on the global test set of the same six classes, without any subsequent task ever being introduced. This setting removes the temporal dimension entirely: there is no class expansion and hence no forgetting, so the only challenges are the decentralized, server-free protocol and the non-IID label skew across the 100 clients. Scenario 1 therefore acts as a controlled sanity check for RQ1—it verifies that CANDLE’s age-based local learner and context-aware peer aggregation can converge to an accurate detector on a fixed data distribution before any continual-learning stress is applied, and it establishes the reference accuracy against which the streaming setting is compared.

Scenario 2 (streaming). The second scenario reproduces the operational condition that motivates the paper. Starting from

Table 3: Model and training configuration.

Component	Setting
Clients	100
Tasks / classes	6 tasks (6/6/6/6/6/4), 34 classes
Partition	non-IID Dirichlet, $\alpha = 5.0$
Backbone	CNN-GRU (DeepFed), see Table 4
Rounds per task	20
Local epochs / round	1
Batch size	2048
Learning rate	10^{-3}

the six base classes of task $t=0$, the five incremental tasks arrive one after another, $t=1 \rightarrow 5$, each revealing a disjoint group of previously unseen attack classes (6, 6, 6, 6, and 4 classes, respectively, for 34 classes in total). A client trains on the current task using only the data available at that stage, with no access to the raw traffic of earlier tasks, and after finishing each task the model is evaluated on the *cumulative* test set covering all classes seen so far (from $t=0$ up to the current task). Because the evaluation pool keeps growing while old data is never revisited, a method must simultaneously acquire the newly introduced classes and retain competence on the earlier ones. This scenario directly targets RQ2: it exposes catastrophic forgetting under a strict server-free, replay-free regime and measures how gracefully each method preserves earlier attack knowledge as the task sequence lengthens.

5.4. Experimental Settings

5.4.1. Experimental Environment

All methods are implemented in PyTorch and trained on GPU accelerators under an identical protocol: the 100 clients train for 20 rounds per task with one local epoch per round, a batch size of 2048, and a learning rate of 10^{-3} . Every method uses the same client partition and the same held-out global test set. Table 3 summarizes the full configuration.

5.4.2. DL-based Classifiers

Every client detects intrusions with a hybrid CNN-GRU classifier adapted from the DeepFed backbone [27], which pairs a convolutional feature extractor with a recurrent branch to capture, respectively, local and sequential patterns in a flow feature vector. Each record in CICIOT2023 is a 39-dimensional feature vector, presented to the network as a length-39 single-channel sequence. The convolutional path stacks three Conv1d-BatchNorm-ReLU-MaxPool blocks with 64, 128, and 256 filters (kernel size 3, “same” padding); each pooling stage halves the temporal length, producing a 256×4 map that is flattened to a 1024-dimensional vector. In parallel, a two-layer GRU with 100 hidden units reads the same input sequence and returns its last hidden state (100 dimensions). The convolutional and recurrent representations are concatenated (1124 dimensions) and passed through a fully connected layer of size 256, a dropout layer ($p=0.5$), and a final 34-way linear layer whose softmax yields the class posterior. The complete architecture, with per-layer output shapes and parameter counts, is given in Table 4; the model has roughly 0.51M trainable parameters. CANDLE and all baselines share this back-

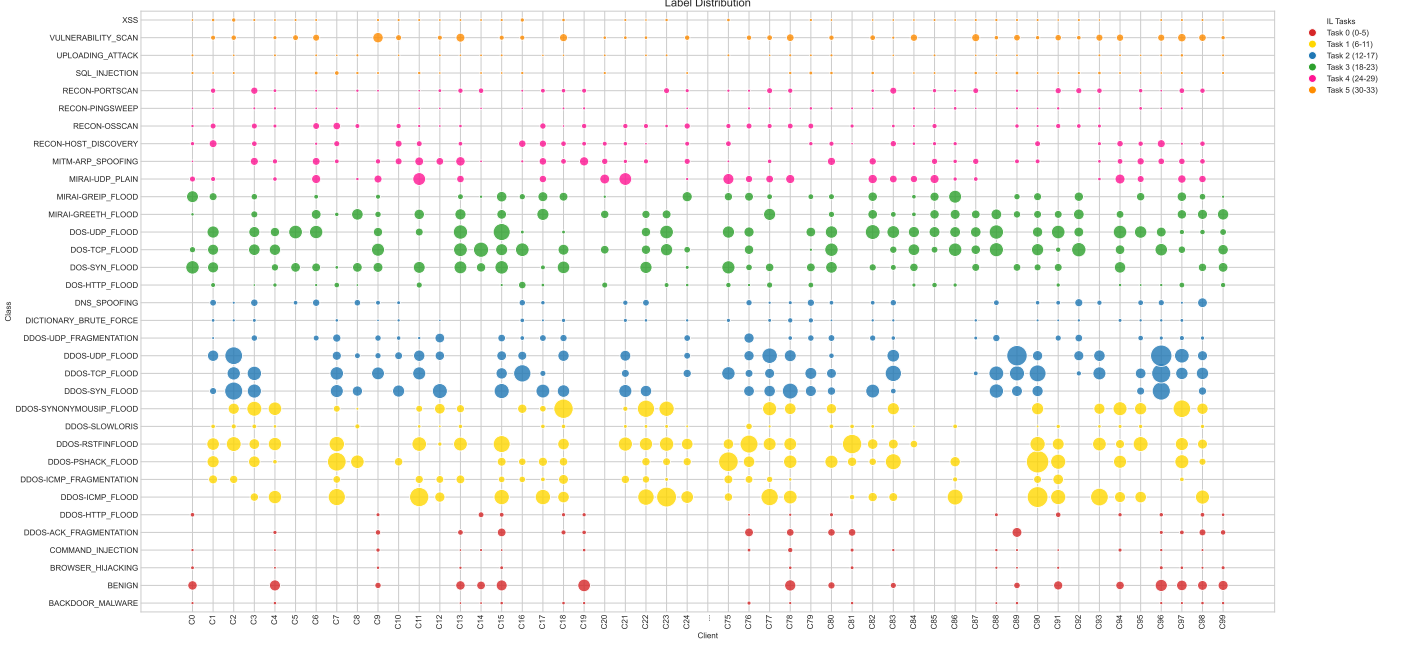


Figure 2: Per-client label distribution for the 100-client non-IID partition ($\alpha = 5.0$). Bubble area is proportional to the number of samples of a given class held by a client.

Table 4: Layer-wise configuration of the CNN-GRU classifier for a 39-feature input and 34 output classes. Output shapes omit the batch dimension.

Layer	Output shape	Param #	Connected to
Input	(39, 1)	0	—
Conv1D-1	(39, 64)	256	Input
BatchNorm-1	(39, 64)	128	Conv1D-1
MaxPool1D-1	(19, 64)	0	BatchNorm-1
Conv1D-2	(19, 128)	24,704	MaxPool1D-1
BatchNorm-2	(19, 128)	256	Conv1D-2
MaxPool1D-2	(9, 128)	0	BatchNorm-2
Conv1D-3	(9, 256)	98,560	MaxPool1D-2
BatchNorm-3	(9, 256)	512	Conv1D-3
MaxPool1D-3	(4, 256)	0	BatchNorm-3
Flatten	(1024)	0	MaxPool1D-3
GRU (2 layers)	(100)	91,500	Input
Concatenate	(1124)	0	Flatten, GRU
Dense-1	(256)	288,000	Concatenate
Dropout	(256)	0	Dense-1
Dense-2	(34)	8,738	Dropout
Softmax	(34)	0	Dense-2
Total		512,654	

bone, so performance differences are attributable to the learning strategy rather than to model capacity.

We compare against three families of methods: (i) server-based federated continual learners, including federated NICE, FedCBDR, CGoFed, FedLwF, FedAvg+EWC [16], and FedAvg+DER [17]; (ii) local (non-collaborative) incremental learners NICE [23], DER [17], LwF, and EWC [16]; and (iii) the server-free decentralized method Plexus. Only Plexus and CANDLE operate without any central coordinator; the federated baselines additionally rely on a server, and several (FedCBDR, DER) retain an exemplar buffer, so they act as an upper reference rather than a like-for-like comparison.

5.4.3. Evaluation Metrics

We assess every method with four standard classification metrics computed on the cumulative test set, plus one CANDLE-specific diagnostic. Let C be the number of classes seen so far, N the number of test samples, and TP_c , FP_c , FN_c the true positives, false positives, and false negatives of class c .

Accuracy (Acc) is the fraction of correctly classified samples:

$$\text{Acc} = \frac{\sum_{c=1}^C TP_c}{N}. \quad (27)$$

Precision and recall of class c , and their harmonic mean, the per-class F1 score, are

$$\text{Pre}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Rec}_c = \frac{TP_c}{TP_c + FN_c}, \quad \text{F1}_c = \frac{2 \text{Pre}_c \text{Rec}_c}{\text{Pre}_c + \text{Rec}_c}. \quad (28)$$

Because the class distribution is highly imbalanced (Fig. 2), we summarize the per-class scores in two ways: the macro F1 (mF1) weights every class equally, whereas the weighted F1 (wF1) weights each class by its support n_c ,

$$\text{mF1} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c, \quad \text{wF1} = \sum_{c=1}^C \frac{n_c}{N} \text{F1}_c. \quad (29)$$

Under class imbalance, mF1 is the more demanding metric because rare attack classes contribute as much as frequent ones. Finally, for CANDLE we report the context-router accuracy, the fraction of samples routed to the correct task/context head, as an internal diagnostic of the routing mechanism. All results are single-run measurements taken at the final round of each task.

5.5. Experimental Results

5.5.1. Scenario 1: Stationary Detection

On the stationary task ($t=0$), CANDLE reaches 74.84% accuracy and 71.46% wF1 at the final round, peaking at 80.34% accuracy earlier in training. This confirms (RQ1) that the decentralized age-based learner, together with context-aware peer aggregation, converges to a usable detector on a fixed distribution. The absolute accuracy is nonetheless below the near-saturated 98–99% that the server-based baselines attain on this single task, indicating that removing the central coordinator and operating under a strict no-replay constraint costs early-task accuracy—a gap we analyze further below.

5.5.2. Scenario 2: Streaming Class-Incremental Detection

Table 5 reports the final-round performance averaged over the six tasks, and Table 6 traces CANDLE task by task alongside the server-free baseline Plexus. Figures 3 and 4 plot the accuracy and weighted F1 of representative methods at every communication round across the six-task stream, making the per-method learning and forgetting dynamics directly visible.

In the server-free setting that matches CANDLE’s assumptions, CANDLE attains the best final-task retention. On the cumulative test set after the last task it keeps 22.49% accuracy, compared with 0.79% for Plexus and 2.67–2.68% for the regularization-based learners (FedAvg+EWC, Local EWC, Local LwF), which effectively collapse to the newest classes. Table 6 shows the mechanism: Plexus starts higher on the early tasks but loses almost all earlier knowledge as new tasks arrive, whereas CANDLE degrades gradually because its mature neurons are shielded from peer aggregation. Averaged over all tasks, CANDLE (43.45% accuracy) is on par with Plexus (43.32%) and the federated regularization baseline FedAvg+EWC (43.08%).

The centralized replay-based methods remain ahead in absolute terms: FedCBDR and federated NICE reach 61.13% and 67.90% accuracy on the final task, respectively, by using a central server and, in the case of FedCBDR, an exemplar buffer. CANDLE deliberately forgoes both, so this gap reflects the cost of the server-free, replay-free regime rather than a like-for-like deficit.

Finally, Table 6 exposes the current bottleneck. CANDLE’s end-to-end accuracy tracks its context-router accuracy closely: routing is perfect on task 0 (100%) but falls to 62.75% by task 2 and to 35.82% by task 5. Because samples routed to the wrong context head cannot be classified correctly, router degradation, rather than the local continual learner, caps performance on later tasks. Improving the router under a growing number of contexts is therefore the primary avenue for closing the gap to the centralized baselines.

5.6. Communication Cost

We characterize communication cost by the number of transmitted bytes per participant and per communication cycle, including both Context Capsules and masked model updates. For CANDLE, this cost depends on the size of the selected neighborhood, the number of trainable coordinates, the representation dimension used by the capsule, and the update encoding.

The experimental comparison should report the total transmitted volume, the mean volume per participant, and the maximum participant load for every evaluated decentralized baseline. Until these measurements are available, we do not draw a quantitative conclusion regarding communication efficiency.

6. Conclusion

This paper has presented CANDLE, a *Capacity-Aware Neurogenesis for Decentralized federated class-incremental learning* framework that jointly addresses three requirements left unresolved by existing federated intrusion detection systems: (i) fully decentralized operation without a coordinating server, (ii) class-incremental accommodation of an expanding attack taxonomy without sharing or reconstructing raw traffic, and (iii) collaborator selection driven by what each participant has actually learned rather than by a static data or gradient signature.

6.1. Summary of Contributions

CANDLE introduces four tightly integrated mechanisms. *First*, an age-partitioned neurogenesis backbone divides every collaborator’s parameters into reserve, young, and mature pools, so that old-class knowledge is protected structurally—by freezing mature neurons and masking their gradients—rather than penalized uniformly as in EWC [16]. Unlike DEN [22] and NICE [23], which were not designed for multi-client aggregation, CANDLE specifies exactly how age-partitioned parameters should be combined when collaborators expand and consolidate their networks asynchronously. *Second*, a lightweight Context Capsule κ_n^t (Eq. (16)), bounded in size by the number of known classes and the parameter count rather than the dataset size, replaces both raw data exchange and the full EWC importance vector as the sole artifact transmitted between peers, preserving the privacy guarantees that motivated the federated setting in the first place. *Third*, a context-aware, age-masked aggregation rule (Eqs. (17)–(19)) restricts peer influence to context-compatible neighbors and to the plastic region of the receiving model, ensuring that a peer whose current task is unrelated never overwrites the receiver’s consolidated knowledge. *Fourth*, a Capacity-Aware Neurogenesis Controller (CANC, Eq. (21)) monitors reserve utilization and prototype drift to decide whether to reuse, expand, or recycle neuron capacity, preventing the capacity exhaustion that would otherwise stall unmanaged neurogenesis over long task sequences—the failure mode left unaddressed by DEN and NICE in isolation.

6.2. Analysis and Discussion

From the methodological perspective, CANDLE departs from the existing literature in several respects. *Compared with centralized FCIL methods* such as GLFC [18], TARGET [19], and FL-IIDS [20], CANDLE eliminates the single point of failure inherent in server-coordinated replay, reconstruction, or distillation. The Context Capsule carries enough structural information—class-wise activation prototypes, age masks, importance scores, and a capacity/reliability scalar—to enable decentralized neighbor selection without requiring a proxy server

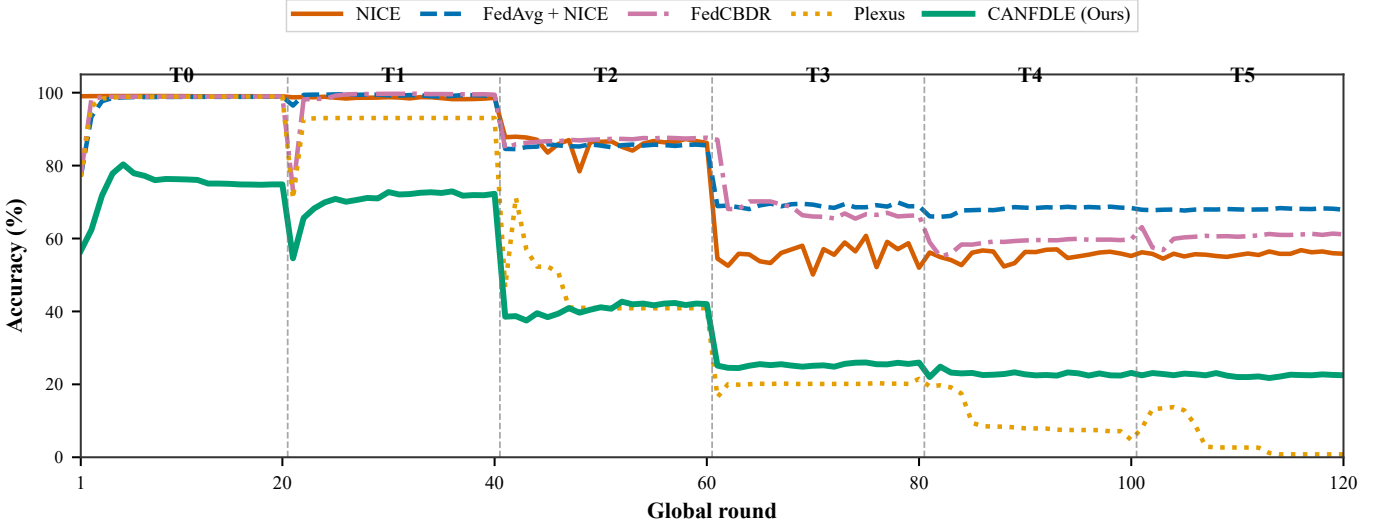


Figure 3: Accuracy at every global communication round for representative methods over the six-task stream. Dashed vertical lines mark the task boundaries T0–T5 (20 rounds per task). Server-based methods (FedAvg+NICE, FedCBDR) stay highest, while among the server-free methods CANDLE holds a higher curve than Plexus on the later tasks.

Table 5: Streaming class-incremental results on CICIOT2023 (100 clients, six tasks). Values are final-round metrics averaged over all tasks; “Task 5 acc.” is the accuracy on the cumulative test set after the last task. “Server-free” marks methods that operate without a central coordinator. In each column, **bold**, underline, and *italic* mark the best, second, and third value.

Method	Setting	Avg. acc.	Avg. wF1	Avg. mF1	Task 5 acc.
Federated NICE	server, replay-free	81.45	80.91	<u>27.64</u>	67.90
FedCBDR	server, replay	<u>78.88</u>	<u>75.23</u>	52.60	<u>61.13</u>
CGoFed	server, replay-free	69.90	62.43	24.01	48.54
FedLwF	server, replay-free	48.10	38.99	11.75	17.84
FedAvg+EWC	server, replay-free	43.08	37.54	20.75	2.67
FedAvg+DER	server, replay	27.44	16.95	7.27	17.84
Local NICE	local, replay-free	<i>74.48</i>	<i>71.06</i>	<i>27.38</i>	<i>55.83</i>
Local DER	local, replay	55.10	54.27	17.60	2.64
Local LwF	local, replay-free	49.21	43.79	9.71	2.68
Local EWC	local, replay-free	43.39	39.14	7.65	2.68
Plexus	server-free	43.32	37.81	20.35	0.79
CANDLE (ours)	server-free	43.45	37.03	18.58	22.49

Table 6: Per-task accuracy on the cumulative test set for the two server-free methods. CANDLE degrades gradually and retains competence on the final task, whereas Plexus collapses. The last column reports CANDLE’s context-router accuracy. **Bold** marks the higher accuracy per task.

Task	Plexus acc.	CANDLE acc.	CANDLE route acc.
$t=0$	98.96	74.84	100.00
$t=1$	93.05	72.26	96.39
$t=2$	40.88	42.01	62.75
$t=3$	21.51	25.97	41.65
$t=4$	4.75	23.14	33.19
$t=5$	0.79	22.49	35.82

to reconstruct old-class data. Compared with decentralized methods such as PENS [10], IFCA [11], NGC [12], and FedCBO [13], CANDLE derives the peer set $\mathcal{N}_n^t$ at every task from a multi-criterion capsule similarity (Eq. (17)) that combines prototype agreement, age-mask overlap, and capacity compatibility, rather than from a single static snapshot such as a validation loss or a one-shot gradient. Compared with XDIFS [25], the closest prior work that also removes the server

and supports streaming classes, CANDLE replaces the uniform EWC penalty across all C parameters with a partition-based formulation in which only the young pool receives gradient updates and only the young pool is exchanged during aggregation. This architectural separation avoids the growing rigidity that arises as EWC accumulates importance scores over many tasks, and it prevents an indiscriminate full-mesh broadcast by conditioning aggregation on capsule-verified compatibility.

The context-aware inference mechanism further distinguishes CANDLE from existing approaches. By routing each test sample through a context detector that matches its binarized activation pattern to the most compatible learned episode, CANDLE achieves task-agnostic inference without requiring an explicit task identifier at deployment time, a practical advantage in streaming network monitoring where the attack family of an incoming flow is unknown a priori.

6.3. Limitations

Several limitations should be acknowledged. First, the current evaluation uses a simulated decentralized environment in

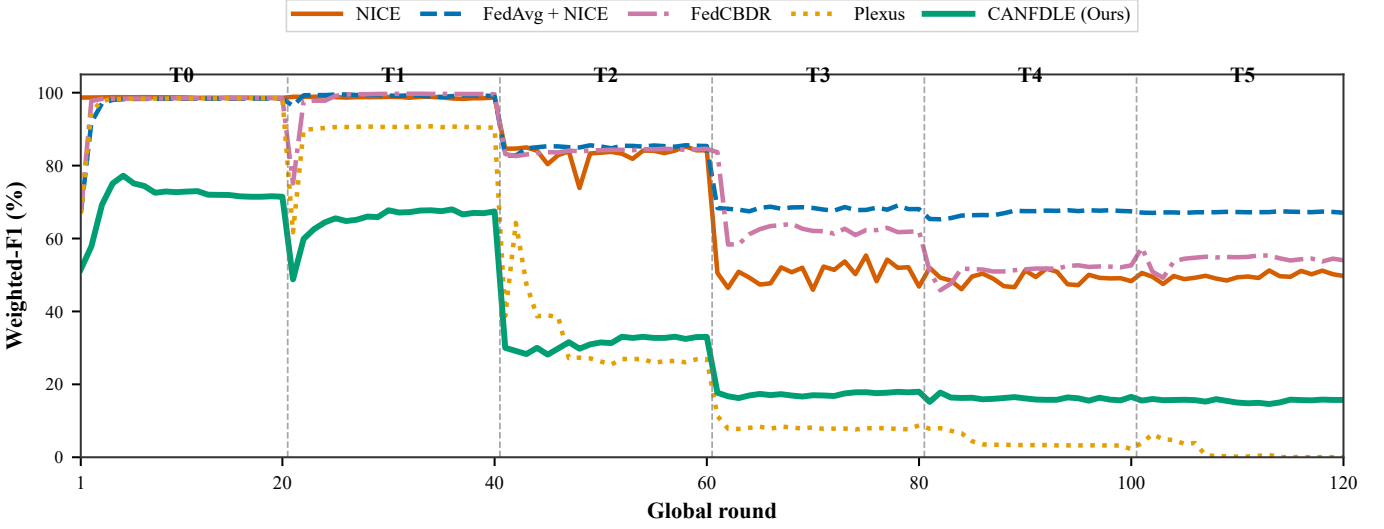


Figure 4: Weighted F1 at every global communication round for representative methods over the six-task stream. Dashed vertical lines mark the task boundaries T0–T5. Plexus collapses more sharply than CANDLE from task T2 onwards, though both trail the server-based baselines.

which all clients execute sequentially on a single compute node; a deployment on a physically distributed testbed with realistic network latency and partial connectivity remains to be validated. *Second*, the capsule broadcast in Step 4 still assumes that every collaborator can reach every other collaborator within a bounded number of hops. In a very large federation (e.g., hundreds of organizations), this all-to-all capsule exchange may become a communication bottleneck, and a gossip-based or hierarchical capsule propagation strategy would need to be investigated. *Third*, the CANCE decision thresholds (θ_1 , θ_2 , θ_3) and the similarity weight vector λ in Eq. (17) are currently set by grid search; an adaptive or meta-learned calibration would improve robustness across heterogeneous deployment conditions. *Fourth*, CANDLE does not currently detect truly novel, previously unseen attack classes autonomously; unknown traffic must be labeled by a Security Operations Center (SOC) analyst before it can be incorporated as a new incremental task. Integrating an open-set or out-of-distribution detection module would close this gap.

6.4. Future Works

Building on the architectural foundation laid by CANDLE, we identify five concrete directions for future work.

1. **Scalable capsule dissemination.** Replace the bounded-hop broadcast of Step 4 with a gossip-based protocol that propagates capsules along a sparse, dynamically rewired overlay graph, reducing per-round communication from $O(N(N-1))$ to $O(N \log N)$ while preserving context-aware neighbor discovery.
2. **Adaptive threshold calibration.** Learn the CANCE thresholds (θ_1 , θ_2 , θ_3), the similarity weights λ , and the capsule-similarity threshold ξ jointly via a lightweight meta-learning or Bayesian optimization loop that adapts to the local data distribution of each collaborator without centralized coordination.

3. **Open-set novelty detection.** Augment the context detector with an energy-based or prototype-distance novelty score so that truly unknown attack families can be flagged automatically, triggering a new incremental task without waiting for manual SOC labeling. This would move CANDLE from a semi-automated to a fully autonomous incremental learning pipeline.
4. **Heterogeneous model architectures.** Extend the capsule exchange and aggregation protocol to support collaborators with different backbone architectures (e.g., varying numbers of convolutional layers or GRU hidden dimensions), enabling cross-architecture knowledge transfer via prototype-level or logit-level fusion rather than parameter-level averaging.
5. **Byzantine-robust aggregation.** Strengthen the age-aware aggregation step with certified Byzantine resilience guarantees, for instance by combining the coordinate-wise median mode already supported (Section 4.3, Step 5) with capsule-based reputation scoring, so that a malicious collaborator injecting poisoned updates is excluded not only by the similarity threshold of Eq. (18) but also by a provably robust aggregation rule.

Acknowledgments

This research was supported by The VNUHCM-University of Information Technology’s Scientific Research Support Fund.

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